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Application and Challenges of Artificial Intelligence in Quadruped Robots

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***Abstract*-This article focuses on the application of artificial intelligence in quadruped robots, and the main achievements are as follows. Optimization of reinforcement learning algorithms: It makes up for the lack of intelligence in traditional control methods. Bionic and bio-inspired strategies: Enhance the natural motion characteristics of the robot. Multimodal perception fusion and intelligent navigation technologies: With the help of technologies such as thermal field path planning and edge computing, it promotes the improvement of the autonomous capabilities of quadruped robots. Problems and challenges: There is still room for improvement in aspects such as the transfer efficiency between the simulation and real environments, the real-time computing ability under the dynamic constraints of hardware, as well as the safety and robustness in complex scenarios. This article aims to provide technical directions and theoretical references for the future research on quadruped robots.**

Keywords-Artificial intelligence; Quadruped robots; Reinforcement learning; Neural networks

I. Introduction

In the field of robotics, wheeled and tracked robots have limitations when facing complex terrains[1]. In contrast, quadruped robots have emerged prominently with their greater flexibility and adaptability. A series of achievements from companies and institutions such as Boston Dynamics, ETH Zurich, and Yushu Technology in China have kept the research enthusiasm for quadruped robots high[2, 3]. Although artificial intelligence has promoted the development of quadruped robots from traditional gait control to intelligent monitoring[4, 5], significantly enhancing their dynamic task execution capabilities, the development of artificial intelligence still faces numerous difficulties in aspects such as simulation - to - reality transfer[6, 7], hardware - algorithm collaboration[8], and safety ethics[9]. Existing relevant reviews have gaps. This article systematically constructs an integration framework of artificial intelligence in quadruped robots for the first time, deeply analyzes various technical challenges, and prospectively maps out a future research roadmap, aiming to fill the key gaps between theory and practice and facilitate the wide application of quadruped robots in practical scenarios such as wild rescue, Industry 4.0, and smart cities.

II. Key Technologies Of Artificial Intelligence In Quadruped Robots

A. Comparison of Typical Quadruped Robot Control Systems

Quadruped robots have garnered significant attention due to their high payload capacity, superior stability, and exceptional adaptability to complex environments. Leading research institutions in quadruped robotics include Boston Dynamics (USA) and Unitree Robotics (China) [2, 3, 10]. The most prominent quadruped robot platforms—ANYmal and Boston Dynamics Spot—differ in control strategies, computing architectures, AI capability and performance index. To gain insights into their technological advancements, we compare these systems in Table 1,Table 2,Table 3 and Table 4.

Table 1 Comparison of Control Strategies

System	Control Strategy	Control Strategy	Control Strategy
ANYmal	Adaptive gait control based on reinforcement learning (RL), combined with model predictive control (MPC) for dynamic stability optimization.	Enables adaptive gait generation for complex terrains (e.g., rough surfaces, stairs) with dynamic balancing and fall recovery capabilities.	Requires extensive training with both simulated and real-world data, leading to long training cycles.
Boston Dynamics Spot	Hierarchical control architecture (high-level task planning + low-level dynamic balancing) with a behavior library and real-time adjustments.	Commercially mature system supporting multi-task expansion (e.g., inspection, material handling); strong self-recovery capabilities after falls.	Closed-source system with limited customization and high hardware maintenance costs.

Table 2 Comparison of Computational Frameworks

System	Computational Architecture	Hardware Configuration	Real-Time Performance & Efficiency
ANYmal	Edge-end hybrid architecture: - End-side: NVIDIA Jetson AGX Xavier (real-time control) - Edge-side: Cloud-assisted computing (complex task planning).	High-performance GPU enables reinforcement learning inference, while edge computing reduces latency (50–100ms).	Balances computational power and energy efficiency, suitable for multi-task processing in dynamic environments.

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Boston Dynamics Spot	Layered computing architecture: - End-side: Custom processor (real-time control). - Edge-side: Onboard computer (task planning and sensor fusion).	Modular design supports hardware expansion (e.g., additional LiDAR), with end-side latency <30ms and edge computing power enabling multi-sensor fusion.	Ensures commercial reliability but operates within a closed hardware ecosystem, making third-party algorithm integration challenging.

TABLE 3 COMPARISON OF AI CAPABILITY

Characteristics	ANYmal	Spot
Reinforcement Learning Algorithm	Uses Deep Reinforcement Learning (DRL) for motion control and complex task learning	Uses model - based control and reinforcement learning for motion optimization and task execution
Perceptual Modalities	Equipped with multi - modal sensors (such as RGB - D cameras, LiDAR, IMU, force/torque sensors)	Equipped with stereo cameras, depth cameras, IMU, force/torque sensors
Autonomous Navigation	Supports SLAM - based autonomous navigation and dynamic obstacle avoidance	Supports map - based autonomous navigation and dynamic obstacle avoidance
Environmental Perception	Able to recognize and classify objects and terrains in complex environments	Able to recognize terrain features and walk adaptively
Task Learning	Supports online learning and transfer learning to adapt to new tasks and environments	Supports pre - programmed tasks and has limited learning capabilities

TABLE 4 COMPARISON OF PERFORMANCE INDEX

Characteristics	ANYmal	Spot
Payload Capacity	Can carry a load of up to 10 - 15 kg	Can carry a load of up to 14 kg
Terrain Adaptability	Able to adapt to complex terrains (such as stairs, gravel, slopes) and dynamic environments	Able to adapt to various terrains (such as grasslands, sand, stairs), but has limited performance in extreme terrains
Battery Life	Approximately 2 - 4 hours (depending on the task and load)	Approximately 1.5 - 2 hours (depending on the task and load)
Movement Speed	Maximum speed of approximately 1.5 m/s	Maximum speed of approximately 1.6 m/s
Ingress Protection Rating	IP67 (can work in harsh environments)	IP54 (dust - proof and waterproof, but not suitable for extreme environments)

B. Optimization and Innovation of Reinforcement Learning Algorithms

Traditional quadruped robot control methods, such as the Zero Moment Point (ZMP) and the Spring-Loaded Inverted Pendulum (SLIP) model, suffer from limited intelligence and low automation levels. In recent years, optimizing and innovating reinforcement learning (RL) algorithms has become a key approach to enhancing quadruped robot motion control.To overcome the deficiencies of traditional RL algorithms, many studies have achieved beneficial improvements by improving the model structure and learning mechanism. For example, in response to the problem of overestimation in the Deep Deterministic Policy Gradient (DDPG) algorithm, Wang et al. [11] adopted a double Q-learning structure to decouple action selection and value estimation, and optimized gait learning, enabling better motion stability in complex environments. In the experiment based on the MIT Cheetah robot, the robot using the double Q-learning structure was able to conduct obstacle-crossing tests at a speed of 90 km/h, and the probability of successfully crossing obstacles increased from 60% to 85%, greatly improving the motion stability compared with the robot without the double Q-learning structure; Kubacki et al. [12] started from the perspective of the reward structure and found that focusing the reward mechanism on behavior learning (rather than the final goal achievement) would greatly promote learning. They also reduced the training cost by reducing the hidden layers of the neural network and adjusting the discount factor γ, and the maximum number of steps of the robot was reduced by up to 93.8%; Jiang et al. [13] utilized the integration of a Spiking Neural Network (SNN) and a policy network to solve the problem of low response time of the traditional Artificial Neural Network (ANN). The low consumption of the SNN increased the lifespan of the robot, and at the same time, greatly improved the control response speed. It reduced the power consumption by 96% compared with the traditional ANN. For the target speeds of 1%, 01%, 01%, 01%, 01% in complex terrains, the deviation range was all controlled within 17%. In the actual test process, when the A1 robot ran continuously for 2 hours, the traditional ANN consumed 500 joules, while the algorithm combining the SNN and the policy network reduced the energy consumption of the A1 robot to 20 joules, fully demonstrating its characteristic of low energy consumption.

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In terms of the research on algorithm generalization and multi-scenario adaptability. Di et al. [14] designed a Meta-Reinforcement Learning (Meta-RL) framework, combined with a Gated Recurrent Unit (GRU), to achieve cross-robot platform strategy transfer. It only required 164 million samples for learning, which greatly reduced the training complexity of the algorithm compared with the 800 million samples required by Feng et al. [15], and also reduced the number of fine-tuning of the robot strategy for new environments. In the experiments on four different types of quadruped robots of multiple brands, the robot using the Meta-RL framework could run stably after about 5 times of fine-tuning, while the robot without this framework needed to be fine-tuned more than 20 times. In terms of the learning efficiency of complex tasks, Shi et al. [16] studied efficient learning based on the world model and the Variational Autoencoder (VAE) framework, and adopted fast fine-tuning from simulation to reality to enable the robot to quickly adapt to dynamic tasks in real scenes. In the task tests on 7 types of terrains (flat ground, slopes, stairs, stones, hills, etc.) in the simulation environment, the policy task completion rate was higher than that of the baseline method. The actual scene test showed that in the simulation experiment of the wild search and rescue scene, the time for the robot to complete the search task in complex terrains was shortened by 30% compared with the baseline algorithm. At the same time, Liu et al. [17] creatively applied Multi-Agent Reinforcement Learning (MARL) to the control of a single robot, regarded each leg as a different agent and established a cooperation mechanism (MASQ framework), which improved the robustness of gait generation and reduced the training cycle. When testing the Boston Dynamics Spot robot, the number of training rounds of the robot using the MASQ framework was reduced to 100, and the gait stability when walking on complex terrains was increased by 25%. The above methods have all promoted the intelligent improvement of reinforcement learning in the control of quadruped robots, and provided algorithm guarantees for the efficient movement of quadruped robots in complex scenes.

C. Motion Control and Adaptation in Complex Environments

In the field of adaptive control for complex terrains, researchers have significantly enhanced the dynamic locomotion capabilities of quadruped robots by deeply integrating biomimetic design with reinforcement learning. Li et al. [18] inspired by biological motion behavior, proposed a training framework based on animal behavior imitation. By optimizing terrain-adaptive movement strategies, their approach improves the robot's natural motion and speed, enabling biological-level agility on rugged surfaces. Similarly, Fei et al [8] addressed gait instability by developing a universal gait control algorithm that leverages AI to analyze the robot's center-of-mass tilt in real time and adaptively adjust gait parameters. This method significantly enhances walking stability across diverse environments. To handle more dynamic scenarios, Li et al. [6] further proposed an intelligent memory-enhanced Soft Actor-Critic (SAC) algorithm. By integrating reinforcement learning with adaptive strategies, their approach enables robots to quickly respond to sudden environmental changes—such as variations in ground material or unexpected obstacles—while maintaining smooth motion transitions. Collectively, these methodologies establish a comprehensive control framework, bridging biological inspiration and intelligent decision-making for quadruped robots.

To meet the diverse motion demands in complex terrains, researchers have developed task-specific algorithms and perception technologies for refined control. Aracting et al. [19] validated the potential of deep reinforcement learning in navigating complex terrains using the Solo 12 robot. Meanwhile, Luo et al. [20] proposed a morphologically adaptive locomotion framework, breaking the limitations of single-robot training by employing a unified control strategy adaptable to robots of different morphologies, thereby reducing deployment costs. In specialized task scenarios, Shi et al. [21] utilized deep reinforcement learning to achieve dynamic object manipulation using quadruped legs, eliminating the need for traditional robotic arms. Bellegarda et al. [22] optimized jump trajectory generation by replacing complex calculations with offset learning and integrating motor dynamic constraints, significantly improving the reliability of jumping motions. Additionally, Sum et al. [23] tackled the risk of slippage during high-speed motion by developing a real-time slip detection system based on convolutional neural networks (CNNs). By fusing multi-sensor data, this system enables precise state perception, providing a fundamental safety assurance for movement in challenging terrains. These technological advancements enhance the environmental adaptability of quadruped robots from multiple dimensions, including motion generation, task execution, and risk prevention, paving the way for intelligent locomotion in complex and dynamic environments.

D. Biomimetic and Bio-Inspired Control

In the realm of biomimetic motion mechanisms and neural system simulation, researchers have advanced the naturalization and adaptability of quadruped robot control by deeply integrating biological principles with intelligent algorithms. Jia et al. [24] pioneered the incorporation of human instinctive responses into robotic control, proposing a hybrid control framework based on electroencephalography (EEG) signals and a central pattern generator (CPG). By coordinating reflex-based control with instinct-driven mechanisms, their approach enables robots to naturally respond to user intentions without requiring complex brain-computer interfaces. Inspired by neuroscience, Schmidgall et al. [25]developed the Synaptic Motion Adaptation (SMA) algorithm, which mimics biological synaptic plasticity and utilizes a three-factor learning rule to enable real-time adaptive adjustments to the environment. Further advancing biomimetic design, Gao et al. [26]combined biomorphic structure and multimodal learning, employing a 2-DoF five-bar linkage mechanism to replicate animal leg morphology. By integrating the GPT-2 language model with visual learning capabilities, their approach endowed robotic dogs with lifelike environmental interactions and behavior evolution characteristics, resembling those of real pets.

In the domains of distributed control and lifelong learning, bio-inspired strategies have demonstrated superior robustness and adaptability. Mortazi et al. [27]introduced the embodiment theory, designing decentralized knowledge-embedded

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actuators to simulate the distributed decision-making mechanisms found in biological organisms. This significantly improved robots' response speed and efficiency in dynamic environments. Addressing the challenges of tendon-driven robot modeling, Marjaninejad et al.[28] proposed a lifelong learning framework based on the G2P algorithm, integrating tactile sensing and neural networks to enable robots to continuously adapt to varying task loads while mitigating catastrophic forgetting. These biomimetic and bio-inspired control paradigms, spanning structural design to learning mechanisms, comprehensively emulate natural biological characteristics. By doing so, they provide innovative solutions for the long-term autonomous operation of quadruped robots in complex environments.

E. Fault Tolerance and Safety Control

To address the mechanical failures and environmental risks that quadruped robots may encounter during complex tasks, researchers have developed multi-level fault-tolerant control and safety assurance strategies. For hardware failures, Hou et al. [29] proposed a proactive fault-tolerant framework based on multi-task learning, which utilizes a symmetrical reflection initialization technique to rapidly reconstruct leg movement patterns. This allows the robot to maintain basic locomotion capabilities even in the event of single-leg or multi-leg failures. Wu et al. [30] further explored actuator degradation scenarios, embedding degradation state information into deep reinforcement learning (DRL) strategies. This enables the control algorithm to dynamically compensate for actuator performance decline, ensuring stable locomotion over long-term operation. Meanwhile, Otomo et al. [9] identified safety vulnerabilities in the joint systems of quadruped robots. Through adversarial attack experiments, they quantified the impact of torque perturbations on locomotion stability, providing critical insights for defensive control design.

In terms of dynamic environmental safety, Zhang et al. [31] proposed a DDPG-driven landing control method for free-fall scenarios. By training on multi-height drop tests, their approach enables the robot's 2-DOF legs to autonomously optimize touchdown posture, significantly reducing impact-induced damage risks. Ji et al. [32] enhanced safety from a perception perspective, designing a multimodal fusion-based end-to-end control architecture that integrates vision, tactile, and inertial data to assess environmental conditions in real-time. Leveraging deep reinforcement learning, their system dynamically adjusts locomotion strategies, achieving fault prediction and proactive obstacle avoidance in complex terrains. These methodologies establish a comprehensive safety framework, spanning fault recovery, anomaly adaptation, and risk prevention, providing technical support for the reliable operation of quadruped robots in extreme scenarios.

F. Enhancing Perception and Navigation Systems

To improve the autonomy of quadruped robots in dynamic environments, researchers have developed multilevel solutions by enhancing multimodal perception and optimizing intelligent navigation. At the perception level, multi-sensor fusion technologies significantly improve environmental understanding. LiDAR provides high-precision 3D point cloud mapping for complex terrain modeling, while RGB-D cameras combine depth information to recognize obstacle materials (e.g., grass and ice). Additionally, IMU and joint encoders offer real-time posture feedback to assist in dynamic balance control.In complex and changeable real-world scenarios, the navigation of quadruped robots faces numerous challenges, and different environmental conditions have a significant impact on their navigation accuracy. Multi-sensor fusion technology has become a key means to improve navigation performance. To clearly present the navigation effects in various environments, Table 5 is compiled to show the sensor fusion methods used under different environmental conditions and the corresponding navigation accuracy indicators.

TABLE 5 NAVIGATION ACCURACY TABLE

Environmental Conditions	Sensor Fusion Method	Navigation Accuracy Indicator	Accuracy Value
Low Light	Fusion of LiDAR and RGB-D cameras (LiDAR provides terrain contours, and RGB-D cameras combine depth information to assist in identifying obstacle materials)	Positioning Error (m)	The positioning error is approximately 0.2 - 0.3 meters in low light
Uneven Surfaces	Fusion of IMU, joint encoders, and LiDAR (IMU and joint encoders provide posture feedback, and LiDAR is used for terrain modeling)	Trajectory Tracking Error (m)	The trajectory tracking error is controlled within 0.15 - 0.25 meters on uneven surfaces
Complex Occlusion Environments	Multi-sensor fusion (LiDAR, vision sensors, etc.) combined with deep learning-based object detection and recognition algorithms	Obstacle Avoidance Success Rate	The obstacle avoidance success rate reaches 90% - 95% in complex occlusion environments

To tackle challenges in sensor data fusion, Youm et al. [33] introduced a Neural Measurement Network (NMN) combined with an Invariant Extended Kalman Filter (IEKF). By leveraging simulation training and domain randomization techniques, their method reduces the sim-to-real gap, achieving up to 50% lower absolute trajectory error. Ji et al. [32] further overcame the limitations of single-modal perception by proposing an end-to-end multimodal fusion strategy, integrating vision, tactile, and inertial data. Using deep reinforcement learning (DRL), their method directly generates motion commands, enabling adaptive gait adjustments in dynamic terrains. For scenarios without external vision inputs, Xue et al. [34] employed Generative Adversarial Imitation Learning (GAIL) combined with Model Predictive Control (MPC). Relying solely on proprioceptive sensors (IMU and joint encoders), their approach enables quadruped robots to navigate complex terrains, demonstrating the robustness of the perception-decision closed loop.

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In navigation and real-time decision-making, coordinated computing architectures and hierarchical strategies play a crucial role. Edge computing (e.g., embedded GPUs) handles high-speed tasks such as gait control, while edge nodes perform multi-sensor fusion and dynamic obstacle avoidance. Meanwhile, cloud computing is responsible for global path planning. Oliveira et al. [35] developed an AI-based navigation system that integrates Bézier curves with dynamic obstacle recognition. Their multi-layered perception neural network predicts obstacle trajectories in real time and generates smooth avoidance paths. Lee et al. [36] introduced an artificial heat field model, simulating thermal conduction to quantify terrain travers ability. By combining LiDAR and IMU data, their approach optimizes path selection, with simulations demonstrating its effectiveness for arbitrarily shaped P-type and H-type obstacles. Segarra et al. [37]combined graph-based algorithms (e.g., Floyd and Dijkstra) with visual perception, improving global path search efficiency while reducing computational overhead. Vazquez et al. [38] integrated edge computing with Modular Workload Partitioning (MWP) to develop an autono decision-making system for Quadruped Unmanned Ground Vehicles (Q-UGV), enabling fast multi-task switching in dynamic environments. These advancements establish a closed-loop system spanning local perception to global autonomous navigation, laying the foundation for reliable operations in complex scenarios. Future research should further explore lightweight algorithms and hardware co-optimization to address multi-objective conflicts and resource constraints in open environments.

III. Case Study On The Application Of Quadruped Robot Ai Technology

In the field of industrial inspection, the "JuYing Lite3" quadruped robot developed by Hangzhou Deep Robotics Co., Ltd. was piloted at Qiandao Lake General Airport in Jiande, Hangzhou, Zhejiang in February 2025. In the complex airport environment, this robot uses bionic and bio - inspired strategies to simulate the walking postures of animals and maintain smooth movements. Through in - depth analysis of the data collected by detection devices such as visible - light cameras and scanning radars using artificial intelligence ker device, it can accurately perceive environmental changes such as the approach of abnormal personnel and minor damage to perimeter facilities while performing real - time loud - speaking and efficient bird - repelling tasks. When performing inspection tasks, the "JuYing Lite3" uses thermal field path - planning technology to reasonably plan the inspection route, shortening the work that traditionally takes several hours for manual inspection to just half an hour, greatly reducing the pressure of manual inspection and improving the inspection quality and efficiency. This fully demonstrates the advantages of quadruped robots in achieving intelligent and efficient operations in industrial inspection scenarios with the help of artificial intelligence technology.

IV. Future Perspectives

The integration of algorithmic advancements, hardware evolution, and interdisciplinary collaboration will drive quadruped robots toward higher autonomy, bio-inspired adaptability, and multi-scenario generalization. Future systems will combine adaptive control frameworks with biomimetic structures and non-invasive brain-computer interfaces (BCIs) to achieve natural motion and human-robot interaction. The self-improvement and energy efficiency of quadruped robots are also essential. Further adjustments and optimizations in algorithms can significantly aid in the robot's learning and efficiency enhancement. While quadruped robots exhibit excellent performance under safe conditions, ensuring task completion or safe task termination under unsafe or fault conditions is equally critical. Developing multimodal perception-driven defense systems can effectively address multiple fault scenarios. Establishing an interpretable AI ethical framework to ensure controllable human-robot collaboration and to facilitate necessary human intervention in critical situations is also crucial.

Moreover, the development of more application scenarios for quadruped robots, such as rescue operations, industrial applications, and urban services, is indispensable. Current challenges are primarily focused on the fidelity of physical interactions between simulation and reality, real-time hardware limitations, and the optimization of multi-objective conflicts. In the future, quadruped robots will leverage the synergy of algorithms, hardware, and scenarios, evolving from motion control to general intelligent agents. They will become core components in Industry 4.0, smart societies, and extreme explorations.

From the perspective of the transfer between simulation and reality, in order to achieve a more efficient simulation-to-reality transfer, attempts can also be made in the following aspects. Firstly, enhance the accuracy of modeling the real environment. Advanced technologies such as advanced sensors (high-precision 3D scanning technology, environmental monitoring, etc.) can be applied to collect effective information about the real environment and accurately incorporate it into the simulation environment, so as to improve the degree of coincidence between the simulation environment and the real environment. Secondly, develop adaptive transfer learning algorithms in terms of transfer learning algorithms. Such algorithms can automatically adjust the robot's strategies and parameters according to the differences and changes between the real environment and the simulation environment, thus achieving a more efficient transfer. For example, the method based on meta-learning can be used. This method learns the transfer rules in multiple simulation environments according to the robot's transfer patterns in the simulation environment, and the robot can efficiently adapt to the newly emerging real environment based on these transfer rules, and so on. Thirdly, apply the Hardware-in-the-Loop (HIL) simulation technology. For instance, introduce the actual robot hardware into the simulation system, directly test the algorithms and hardware in the real environment, and anticipate in advance various problems that may occur in the practical application of the algorithms and robots in the real world, so as to ensure the operational reliability of the robot in the real environment. The application of these comprehensive measures can significantly improve the efficiency of simulation-to-reality transfer and promote the practical application process of quadruped robots.

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V. Conclusions

Through algorithmic innovation and interdisciplinary integration, artificial intelligence technology has significantly enhanced the motion control, environmental adaptability, and autonomous decision - making capabilities of quadruped robots. The optimization of reinforcement learning algorithms has addressed the generalization limitations of traditional methods, enabling efficient gait generation and cross - scenario transfer. Bionic strategies endow quadruped robots with natural movement and lifelong learning characteristics similar to those of living organisms. Multi - modal perception fusion and intelligent navigation systems have established a closed - loop from local decision - making to global autonomy. In terms of safety mechanisms, fault - tolerant control and dynamic risk prediction have formed a three - dimensional protection system.

In safety-critical disaster response scenarios, the deployment of quadruped robots faces numerous ethical challenges. Their autonomous decision-making may violate the principle of equal rescue due to unequal resource allocation. There is a risk of privacy leakage during data collection and transmission, and excessive reliance on robots may also undermine the sense of responsibility and on-site decision-making ability of human rescue workers. Mitigation strategies include designing transparent and interpretable algorithms and setting up options for human intervention, establishing strict data encryption, access control, and clarifying the regulations for the entire data process. Additionally, it is necessary to emphasize human-robot collaboration, assigning robots to undertake basic search and information collection tasks in dangerous environments, while humans are responsible for core rescue decision-making and the rescue links that require emotional care.

However, the development of quadruped robots still faces several key issues. The simulation - to - reality transfer problem, which seriously restricts the process of quadruped robots moving from the laboratory to the commercial market, ranks first. This problem greatly hinders the effective integration of academic research results and practical application scenarios. The computational efficiency problem follows closely. The current low computational efficiency makes it difficult for robots to meet the stringent requirements of practical application scenarios for rapid response and decision - making, such as in industrial inspection and disaster rescue. It also limits the further exploration and practice of complex algorithms and models in academic research. The Rubler problem ranks third. The lack of Rubler in the commercial application of quadruped robots can lead to frequent malfunctions in complex environments, resulting in increased after - sales costs and damaged brand images. Insufficient robustness also has a negative impact on the credibility and generality of research results in the academic research field.

Based on the analysis of the above - mentioned challenges, this article puts forward the following targeted suggestions for research priorities. First, constructing a hybrid simulation - reality training framework should be given top priority. Customized training strategies should be developed for different application scenarios. By building a system that can provide real - time feedback on real - world data, the gap between simulation and reality can be effectively bridged, providing a solid foundation for the commercialization of quadruped robots and the in - depth development of academic research, thus laying a solid theoretical foundation for the operational implementation of quadruped robots. Second, edge computing optimization should also be a key research direction. This includes developing dedicated edge - computing chips, improving the hardware architecture collaboration mechanism, and exploring distributed computing models to comprehensively improve the data processing efficiency of robots in complex scenarios and meet real - time computing requirements.

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